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A Introduction

Goal of the work, maybe a simple example, some mention to preliminary information and a soft mention to what's the problem and the philosophy.

B Minkowski space and space-like surfaces

{sc:splsrf}

The environment we will work within is the Minkowki 3-space, defined as $\mathbb{R}^3$ endowed with the pseudometric

$$ds^2 = (dx^1)^2 + (dx^2)^2 - (dx^3)^2.$$

In order to distinguish the use of this metric from the standard euclidean metric of $\mathbb{R}^3$ we will use the notation $\mathbb{M}^{2,1}$. We will also denote the coordinates either as (x^1, x^2, x^3) or (x, y, t), choosing the notation according to convenience and will identify the slice $\{(x, y, 0)\}$ of $\mathbb{M}^{2,1}$ with $\mathbb{R}^2$ and with $\mathbb{C}$.

The space $\mathbb{M}^{2,1}$ is an example of a more general object called a pseudo-Euclidean vector space, a term that refers to a finite-dimensional vector space V endowed with a nondegenerate (possibly not positive-definite) symmetric bilinear form g. This kind of space serves as the "infinitesimal model" for pseudo-Riemannian manifolds, smooth manifolds endowed with a smooth nondegenerate 0-2 tensor. If the signature of this tensor is exactly $(n-1, 1)$, the manifold is called a Lorentzian manifold.
The following definition, borrowed from physics, will be useful later.

Definition. Given (M, g) a Lorentzian manifold and $p \in M$, we say that a vector $v \in T_pM$ is

- space-like if $\langle v, v\rangle > 0$,
- light-like if $\langle v, v\rangle = 0$ and $v \neq 0$,
- time-like if $\langle v, v\rangle < 0$.

We also call a subspace $P \subseteq T_pM$ space-like if every nonzero vector in P is space-like.

We will always denote with S a differentiable manifold of dimension 2 (a surface) immersed in $\mathbb{M}^{2,1}$, $\iota : S \hookrightarrow \mathbb{M}^{2,1}$. As usual, the immersion induces a nondegenerate bilinear form on S, the first fundamental form or pull-back pseudometric, which we indicate as $\mathbf{I} = \iota^*(g_{\mathbb{M}^{2,1}})$ and write in coordinates as $\mathbf{I} = g_{ij}dx^i dx^j$ (for simplicity we sometimes forget the immersion ι).

Space-like immersions.
The objects we will focus on are not arbitrary immersed surfaces, but the so-called space-like surfaces of $\mathbb{M}^{2,1}$. This term refers to immersed surfaces in $\mathbb{M}^{2,1}$ whose induced metric is positive-definite, and hence Riemannian. Equivalently, these are the immersed surfaces S such that, for every $p \in S$, the tangent plane T_pS is space-like.

A nice property of this kind of surface is that they are locally the graph of a function of the form $x^3 = f(x^1, x^2)$, f being smooth and Lipschitz with constant less than 1.
We can show this with a simple argument:
First, we assumed that the restriction of the Lorentzian metric to each tangent space T_pS is positive-definite so that it's a space-like plane. This can be rewritten as

$$T_pS \subset \mathscr{C} := \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle \geq 0\} = \{t^2 \leq x^2 + y^2\} \ .$$

The set $\mathscr{C}$ is the outside-region of the double cone in $\mathbb{M}^{2,1}$ whose axis is $\{x^1 = 0,\ x^2 = 0\}$. Clearly $\mathscr{C}$ does not contain any vertical vector and clearly also T_pM doesn't. Therefore we can apply the implicit function theorem and express S locally near p as the graph of a function $t = f(x, y)$, where $f : U \subset \mathbb{R}^2 \to$

$\mathbb{R}$. Where this local representation is valid, we can parametrize the surface as $(x, y, f(x, y))$, and obtain the local frame

$$V_x = (1, 0, \partial_x f(x, y)), \qquad V_y = (0, 1, \partial_y f(x, y)),$$

for which the first fundamental form is represented by the matrix

$$g = \begin{pmatrix} \langle V_x, V_x \rangle & \langle V_x, V_y \rangle \\ \langle V_y, V_x \rangle & \langle V_y, V_y \rangle \end{pmatrix} = \begin{pmatrix} 1 - f_x^2 & -f_x f_y \\ -f_x f_y & 1 - f_y^2 \end{pmatrix}.$$

Since the parametrization is regular, the condition that g be positive-definite is equivalent to

$$0 < \det g = 1 - (f_x^2 + f_y^2) = 1 - |\nabla f|^2,$$

and therefore $|\nabla f|^2 < 1$.

Entire surfaces.
Given a surface S immersed in $\mathbb{M}^{2,1}$, we can define a projection $\Pi : S \to \mathbb{R}^2$ by forgetting the time coordinate, $\Pi(x^1, x^2, x^3) := (x^1, x^2)$ for any $(x^1, x^2, x^3) \in S$. We call the surface S is entire when the map Π is surjective. If the surface is globally a graph of a function φ, this condition is exactly stating that the function φ is an entire map from $\mathbb{R}^2$ to $\mathbb{R}$.

The situation for space-like surfaces is actually a lot nicer than what we presented up to now, as the representation as a graph is not only local but actually global as we prove in the proposition below.

{prop:topproprspimm}

Proposition 1. *Let Ω be a domain in $\mathbb{R}^2$ (that is, an open connected set) and let $\iota : \Omega \hookrightarrow S \subset \mathbb{M}^{2,1}$ be a space-like immersion. Suppose that S is complete with respect to the induced Riemannian metric. Then S is an entire graph and Ω is simply connected.*

Before laying out the proof let's recall a definition and a proposition from the theory of coverings. Let M and N two differentiable manifolds and $f : M \to N$ a smooth map.

Definition B.1. We say that the map f satisfies the path lifting property if for every $p \in N$ and for every path $\alpha : I \to M$ such that $\alpha(0) = f(p)$ there exists a path $\tilde{\alpha} : I \to N$ such that $\tilde{\alpha}(0) = p$ and $f \circ \tilde{\alpha} = \alpha$. We can assume without loss of generality that the paths involved are differentiable.

Proposition 2. *If f is a local homeomorphism and satisfies the path lifting property, then it's a covering map.*

Proof of Proposition 1. The idea of the proof is to show that $\Pi \circ \iota$ is a global diffeomrphism. Clearly this implies that Π restricted to S is a diffeomeorphism on $\mathbb{R}^2$ and so that S is an entire graph. From that we obtain also that Ω is simply connected since it's diffeomorphic to $\mathbb{R}^2$.

Let's observe that the target of $\Pi \circ \iota$ is just $\mathbb{R}^2$, which is simply connected, and the domain is Ω, which is connected since it's a domain. We already proved, by showing that S is locally a graph, that the map $\Pi \circ S$ is a local homeomorphism.

Having met these conditions, if we show that $\Pi \circ \iota$ satisfies the path lifting property, we conclude that it is a smooth covering map whose domain is simply connected, hence a diffeomorphism.

Let's fix a point $z \in \Omega$ and a path $\gamma : [0,1] \to \mathbb{R}^2$ of bounded length, $\ell(\gamma) < \infty$, such that $\gamma(0) = (\Pi \circ \iota)(z)$. Our goal is to show that γ admits a lift to a path γ^* on Ω starting at z.
To this end, let's consider the set of times up to which such a lift exists,

$$\mathcal{J} := \{\tau \in [0,1] \mid \exists \gamma^* : [0,t] \to \Omega \text{ such that } (\Pi \circ \iota) \circ \gamma^* = \gamma \text{ and } \gamma^*(0) = z\},$$

and let's define $\bar{\tau} := \sup \mathcal{J}$.

Let's observe that $0 \in \mathcal{J}$ and that $\mathcal{J}$ is open. This is true because, being $\Pi \circ \iota$ a local homeomorphism, if a certain τ is in $\mathcal{J} \cap (0,1)$ then the whole interval $(\tau - \varepsilon, \tau + \varepsilon)$ for $\varepsilon > 0$ sufficiently small, belongs to it.

If we prove that $\bar{\tau} = 1$, then γ admits a lift on the whole interval $[0,1]$ and we have finished. Then let's suppose by contradiction that $\bar{\tau} < 1$ and define $\gamma^* : [0, \bar{\tau}) \to \Omega$ the lift of γ up to $\bar{\tau}$. This curve exists by definition of $\bar{\tau}$.

Finally, we need to prove that the limit $\lim_{\tau \to \bar{\tau}} \gamma^*(t)$ exists. If this is true we can then extend γ^* up to $\bar{\tau}$, which means that $\bar{\tau} \in \mathcal{J}$. Since $\mathcal{J}$ is open, there exists an element $\bar{\tau} + \varepsilon \in \mathcal{J}$, $\varepsilon > 0$. This contradicts the definition of $\bar{\tau}$ and proves $\bar{\tau} = 1$ since the alternative is absurd.

To prove that $\lim_{\tau \to \bar{\tau}} \gamma^*(\tau)$ exists we just need a couple of observations:
First we note that the map Π dilates distances, as

$$\begin{aligned} \langle v, v \rangle_{\mathbb{M}^{2,1}} &= (v^1)^2 + (v^2)^2 - (v^3)^2 \\ &\leq (v^1)^2 + (v^2)^2 \\ &= \langle \Pi v, \Pi v \rangle_{\mathbb{R}^2} . \end{aligned}$$

From this it's clear that $\ell(\gamma^*) \leq \ell(\gamma)$.
Now, if we fix any sequence $\tau_k \to \bar{\tau}$ and observe that as γ is differentiable, it's derivative is bounded, so $\ell(\gamma|_{[\tau_k, \tau_m]}) \leq C|\tau_k - \tau_m|$. By observing that

$$\begin{aligned} d(\gamma^*(\tau_k), \gamma^*(\tau_m)) &\leq \ell(\gamma^*|_{[\tau_k, \tau_m]}) \\ &\leq \ell(\gamma|_{[\tau_k, \tau_m]}) \\ &\leq C|\tau_k - \tau_m| \end{aligned}$$

we deduce that also $(\gamma^*(\tau_k))_k$ is a Cauchy sequence, hence converging since S is complete and Ω is to be intended with the induced metric.
By the arbitrariness of the sequence $(\tau_k)_k$ we deduce the limit $\lim_{\tau \to \bar{\tau}} \gamma^*(\tau)$ exists. $\square$

In this section we have essentially proved that a complete space-like surface immersed in $\mathbb{M}^{2,1}$ is actually the graph of a smooth function f satisfying $|\nabla f|^2 < 1$. A partial converse also holds: the graph of a smooth function whose gradient has norm strictly smaller than 1 is a space-like surface, although it may lack completeness.

Normal vectors to a space-like surface in $\mathbb{M}^{2,1}$
Since the ambient metric g is nondegenerate and each tangent plane T_pS is space-like, the restriction of g to T_pS is positive-definite, hence nondegenerate. It follows that, for every $p \in S$,

$$\mathbb{M}^{2,1} = T_p\mathbb{M}^{2,1} = (T_pS)^\perp \oplus T_pS \ .$$

Since S has codimension 1 in $\mathbb{M}^{2,1}$, the orthogonal complement $(T_pS)^\perp$ is one-dimensional. Therefore it is generated by a vector $N = N_p$ normal to S.

We can easily show that N must be a time-like vector, since T_pS is space-like.
Let's fix $e_1, e_2 \in T_pM$ two independent vectors. Since N belongs to $(T_pM)^\perp$ we have $\langle e_1, N\rangle = \langle e_2, N\rangle = 0$. This implies that when we look at the representative matrix of g_p written in the basis (N, e_1, e_2), we have

$$[g_p] = \begin{pmatrix} \langle N, N\rangle & 0 & 0 \\ 0 & \langle e_1, e_1\rangle & \langle e_1, e_2\rangle \\ 0 & \langle e_2, e_1\rangle & \langle e_2, e_2\rangle. \end{pmatrix} =: \begin{pmatrix} \langle N, N\rangle & 0 \\ 0 & [(g|_S)_p] \end{pmatrix}.$$

Where the block $[(g|_S)_p]$ represents the form g restricted to the subspace T_pM. We conclude that $\langle N, N\rangle$ is negative just by observing that $\det[g_p] < 0$ as g is a Lorentzian metric, while $\det[(g|_S)_p] > 0$ since g becomes a metric when restricted to S.

Since $(T_pS)^\perp$ is 1-dimensional, all the vectors in it are proportional to each other. Moreover, as we have just seen, they are all timelike, so if w is a nonzero vector in $(T_pS)^\perp$, then

$$\langle w, w\rangle < 0.$$

We can thus rescale w by a factor $k \in \mathbb{R}$ such that $\langle kw, kw\rangle = -1$. This equation yields two possible choices of k, one for each connected component of the space $\hat{\mathbb{H}}^2 = \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle = -1\}$. We can consistently choose the connected component with positive time

$$\mathbb{H} = \mathbb{H}^2 = \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle = -1,\ v^3 > 0\} \ ,$$

effectively selecting a unique normal vector for each $p \in S$. The choice of the component within $\{v^3 > 0\}$ or $\{v^3 < 0\}$ is sometimes referred to a "time-orientation" for the space $\mathbb{M}^{2,1}$, and can be performed independently from the surface S. This way we obtain a globally defined map

$$\tilde{u} : S \to \mathbb{H} \ . \tag{1}$$ {eq:gausshmap}

$\mathbb{H}$ is a model for the Hyperbolic plane.
The space $\mathbb{H}$ that we just defined is clearly a surface immersed in $\mathbb{M}^{2,1}$, but there is more. First of all, we observe that, by manipulating the definition of $\mathbb{H}$, we can write it as the graph of the function $\varphi(x, y) = \sqrt{x^2 + y^2 + 1}$. The gradient of such function is $\nabla\varphi = \frac{1}{\sqrt{x^2+y^2+1}}(x, y)$, from which it's clear that $|\nabla\varphi| < 1$. By applying the result we proved before, we obtain that the bilinear form that $\mathbb{H}$ inherited from the immersion in $\mathbb{M}^{2,1}$ is a Riemannian metric. In

fact this space is one of the many models of the hyperbolic plane.
Another model for the same Riemannian manifold is the Poincaré disk. As a differentiable manifold, the Poincaré disk is simply the unit ball in $\mathbb{R}^2$, $\mathbb{D} = \{\xi \in \mathbb{R}^2 \mid |\xi| \leq 1\}$, while we can construct the metric by exhibiting a diffeomorphism between $\mathbb{D}$ and $\mathbb{H}$ pulling back the metric.
A possible diffeomorphism is the map p, whose inverse is the map q, as can be verified by direct computation:

$$\begin{aligned} p\colon \mathbb{H} &\longrightarrow \mathbb{D} & q\colon \mathbb{D} &\longrightarrow \mathbb{H} \\ (x,y,t) &\longmapsto \frac{1}{1+t}(x,y) & \xi &\longmapsto \frac{1}{1-|\xi|^2}(2\xi^1, 2\xi^2, 1+|\xi|^2). \end{aligned}$$

It is clear that both functions are smooth, since $|\xi| < 1$ and $t > 1$.

We would like to find the expression of $q^*(g_{\mathbb{H}})$.To simplify the calculations we might observe that the metric of $\mathbb{H}$ inherited by $\mathbb{M}^{2,1}$ is invariant under a rotation on the t axis, and that such a rotation corresponds under p to a rotation of the disk. This means that the metric $q^*(g_H)$ will be invariant under rotation. It is then sufficient to fix a certain $\xi \in \mathbb{D}$ of the form $\xi = (a, 0)$ and calculate $q^*(g_{\mathbb{H}})_\xi$; this will specify the whole metric tensor. The matrix associated to dq_ξ in the canonical bases $\mathscr{B}_1 = (e_1, e_2)$ of $\mathbb{R}^2$ and of $\mathscr{B}_2 = (E_1, E_2, E_3)$ of $\mathbb{M}^{2,1}$ is

$$[dq_\xi]_{\mathscr{B}_1\mathscr{B}_2} = \frac{2}{(1-|\xi|^2)^2} \begin{pmatrix} 1+(\xi^1)^2-(\xi^2)^2 & 2\xi^1\xi^2 \\ 2\xi^1\xi^2 & 1-(\xi^1)^2+(\xi^2)^2 \\ 2\xi^1 & 2\xi^2 \end{pmatrix}.$$

Which, evaluated at $\xi = (a, 0)$, yields

$$[dq_{(a,0)}]_{\mathscr{B}_1\mathscr{B}_2} = \frac{2}{1-a^2} \begin{pmatrix} \frac{1+a^2}{1-a^2} & 0 \\ 0 & 1 \\ \frac{2a}{1-a^2} & 0 \end{pmatrix} =: \frac{2}{1-a^2} \left(v_1 | v_2\right).$$

We observe that, with respect to the metric of $\mathbb{M}^{2,1}$, it holds

$$\langle v_1, v_1 \rangle = 1, \qquad \langle v_2, v_2 \rangle = 1, \qquad \langle v_1, v_2 \rangle = 0.$$

Therefore $\mathscr{B}_3 = (v_1, v_2)$ is an orthonormal basis of $T_{q(0,a)}\mathbb{H}$ with respect to $g_{\mathbb{H}}$, hence $[g_{\mathbb{H}}]_{\mathscr{B}_3} = I$, while $dq_{(a,0)}$ represented by

$$[dq_{(a,0)}]_{\mathscr{B}_3\mathscr{B}_1} = \frac{2}{1-a^2} I.$$

So we have

$$\begin{aligned} [q^*(g_{\mathbb{H}})_{(0,a)}]_{\mathscr{B}_1} &= [dq_{(0,a)}]^T_{\mathscr{B}_3\mathscr{B}_1} [g_{\mathbb{H}}]_{\mathscr{B}_3} [dq_{(0,a)}]_{\mathscr{B}_3\mathscr{B}_1} \\ &= [dq_{(0,a)}]^T_{\mathscr{B}_3\mathscr{B}_1} I [dq_{(0,a)}]_{\mathscr{B}_3\mathscr{B}_1} \\ &= \left(\frac{2}{1-a^2}\right)^2 I \quad . \end{aligned}$$

In particular, $q^*(g_{\mathbb{H}})_{(0,a)}$ is a multiple of the Euclidean metric.
By the argument on rotational invariance, we conclude that for every $\xi \in \mathbb{D}$,

$$q^*(g_{\mathbb{H}})_\xi = \left(\frac{2}{1-|\xi|^2}\right)^2 g^E_\xi.$$

The disk $\mathbb{D}$ with this metric, which we denote also with ds_p^2, is called the Poincaré model of the Hyperbolic plane.

The Gauss Map in $\mathbb{M}^{2,1}$
In (1) we constructed a map $\tilde{u}$ between the surface S and $\mathbb{H}$. Having now at our disposal the isometry q between $\mathbb{H}$ and the Poincaré disk, $\mathbb{D}$, we can compose $\tilde{u}$ with q, obtaining what is called the Gauss map of the surface S.

$$u := q \circ \tilde{u} : S \to \mathbb{D}.$$

The construction we followed is analogous to the classical one, except that the target of the Gauss map is here $\mathbb{D}^2$ instead of $\mathbb{S}^2$. Moreover, unlike the Euclidean case, for a space-like surface we always have a globally defined unit normal field: once a time orientation of $\mathbb{M}^{2,1}$ is fixed, the timelike line $(T_pS)^\perp$ intersects the corresponding connected component of $\hat{\mathbb{H}}^2$ in exactly one point for every $p \in S$. Instead, for an immersed surface in Euclidean space, there is no canonical way to choose a normal vector.

C The Gauss-Codazzi equations and the curvatures of S

In this section we derive the Gauss-Codazzi equations for a space-like surface immersed in $\mathbb{M}^{2,1}$, since in this setting they differ in some important respects from the classical case. These equations will play a central role in the discussion that follows.

C.1 Recall on the second fundamental form

We repeat now the construction of the second fundamental form of the immersion ι, which brings a different sign convention that will occur also to the expression of the Gaussian curvature.

As we have already noted, given a point $p \in S$, since S is a space-like surface, we can decompose the tangent space

$$T_p\mathbb{M}^{2,1} = T_pS \oplus (T_pS)^\perp \ .$$

This splitting is differentiable, in the sense that the mappings

$$(p, v) \mapsto (p, v^T) \qquad \text{and} \qquad (p, v) \mapsto (p, v^N)$$

are differentiable maps of $T\mathbb{M}^{2,1}$ to $T\mathbb{M}^{2,1}$.

Denote with D the Levi-Civita connection on $\mathbb{M}^{2,1}$ induced by the pseudometric $g_{\mathbb{M}^{2,1}}$ and with ∇ the connection induced on the surface S. As usual, if X and Y are local vector fields on S with local extensions $\bar{X}$ and $\bar{Y}$ then the induced connection on S is simply the projection of the connection of the ambient space onto the surface S

$$\nabla_X Y = \left(D_{\bar{X}}\bar{Y}\right)^T \ .$$

Which means that the difference between the two connections of X and Y will be a local vector field on $\mathbb{M}^{2,1}$ normal to S,

$$B(X,Y)_p := (D_{\bar{X}}\bar{Y} - \nabla_X Y)_p \perp T_pS. \quad . \tag{2}$$ {eq:defB}

As in the Euclidean case, B is a symmetric tensor and is independent of the chosen extensions $\overline{X}$ and $\overline{Y}$. Since the normal space $(T_pS)^\perp$ is one-dimensional, once a local unit normal vector field N along S is fixed, there exists a unique symmetric bilinear form $\mathbf{II} = \mathbf{II}(X,Y)$ such that

$$B(X,Y)_p = \varepsilon(\mathbf{II}(X,Y)N)_p,$$

where we can choose $\varepsilon = \pm 1$. Let's find a suitable choice for ε.
From now on we forget to distinguish a vector field X with its extension $\overline{X}$. Let N be local normal vector field on S and V, W two local tangent vector fields to S. We observe first that

$$\langle N, W\rangle \equiv 0 \ ,$$

so that by taking the derivative along V,

$$\langle D_V N, W\rangle + \langle N, D_V W\rangle = 0 \quad .$$

Then

$$\begin{aligned}\langle D_V N, W\rangle &= -\langle N, B(V,W)N + \nabla_V W\rangle \\ &= -\langle N, B(V,W)N\rangle \\ &= -\langle N, \varepsilon\mathbf{II}(V,W)N\rangle \\ &= -\varepsilon\langle N, N\rangle\, \mathbf{II}(V,W) \\ &= \varepsilon\mathbf{II}(V,W) \quad .\end{aligned}$$

So the natural choice is $\varepsilon = -1$.
With this choice we have, from definition 2,

$$D_V X = \nabla_V X + \mathbf{II}(V,X)N \quad . \tag{3}$$ {eq:gc1}

C.2 The Gauss-Codazzi equations and Gaussian curvature

It's easy to see that $\mathbb{M}^{2,1}$ is actually flat with its metric. This mean that its curvature tensor is zero $\overline{R}(X,Y)Z \equiv 0$. Let's rewrite $\overline{R}$ in relation with R the curvature tensor of the surface S. By definition

$$\overline{R}(X,Y)Z = D_X D_Y Z - D_Y D_X Z - D_{[X,Y]}Z \quad .$$

Let's focus first on the term $D_X D_Y Z$, applying the equation (3) we get

$$\begin{aligned}D_X D_Y Z &= D_X(\nabla_Y Z + \mathbf{II}(Z,Y)N) \\ &= \nabla_X\nabla_Y Z + \mathbf{II}\left(\nabla_Y, X\right)N + d_X(\mathbf{II}(Z,Y))N + \mathbf{II}(Z,Y)D_X N \quad .\end{aligned}$$

And analogously

$$D_Y D_X Z = \nabla_Y\nabla_X Z + \mathbf{II}\left(\nabla_X, Y\right)N + d_Y(\mathbf{II}(Z,X))N + \mathbf{II}(Z,X)D_Y N \quad .$$

Finally we have, by noting that the connection ∇ is symmetric,

$$\begin{aligned} D_{[X,Y]}Z &= \nabla_{[X,Y]}Z - \mathbf{II}([X,Y],Z)N \\ &= \nabla_{[X,Y]}Z + \mathbf{II}(Z,\nabla_X Y)N - \mathbf{II}(Z,\nabla_Y X)N \quad . \end{aligned}$$

So that, by putting everything together we obtain,

$$\begin{aligned} 0 &= \overline{R}(X,Y)Z \\ &= D_X D_Y Z - D_Y D_X Z - D_{[X,Y]}Z \\ &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]}Z \\ &\qquad + d_X(\mathbf{II}(Z,Y))N - \mathbf{II}(\nabla_X Z,Y)N - \mathbf{II}(Z,\nabla_X Y)N \\ &\qquad - [d_Y(\mathbf{II}(Z,X))N - \mathbf{II}(\nabla_Y Z,X)N - \mathbf{II}(Z,\nabla_Y X)N] \\ &\qquad + \mathbf{II}(Z,Y)D_X N - \mathbf{II}(Z,X)D_Y N \quad . \end{aligned} \tag{4}$$

{eq:gc2}

Now, we recognize in the expression that $R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]}Z$, and by definition, we set the derivative of the tensor,

$$(\nabla_X(\mathbf{II}))(Y,Z) = d_X(\mathbf{II}(Z,Y)) - \mathbf{II}(\nabla_X Z,Y) - \mathbf{II}(Z,\nabla_X Y) \ .$$

So, by continuing in (4),

$$\begin{aligned} 0 = R(X,Y)Z + (\nabla_X \mathbf{II})(Y,Z)N - (\nabla_Y \mathbf{II})(Z,X)N \\ + \mathbf{II}(Z,Y)D_X N - \mathbf{II}(Z,X)D_Y N \quad . \end{aligned} \tag{5}$$

{eq:gc3}

Now we observe that from $\langle N,N\rangle \equiv -1$ we have, for v tangent to S,

$$2\langle D_v N, N\rangle = 0$$

so that $D_v N$ is a tangent vector on S. Now, in (5) we can separate the terms that are tangent to S and those that are normal. This way we obtain two distinct equations

$$\begin{cases} R(X,Y)Z = -\mathbf{II}(Z,Y)D_X N + \mathbf{II}(Z,X)D_Y N \\ (\nabla_X \mathbf{II})(Y,Z) - (\nabla_Y \mathbf{II})(Z,X) = 0 \end{cases} \quad . \tag{6}$$

{eq:gc4}

These equations are (a form of) the Gauss-Codazzi equations.

Let's take the first equation of (5) and test it by a tangent local vector field W. We get

$$\begin{aligned} \langle R(X,Y)Z,W\rangle &= \langle -\mathbf{II}(Z,Y)D_X N + \mathbf{II}(Z,X)D_Y N, W\rangle \\ &= -\mathbf{II}(Z,Y)\langle D_X N, W\rangle + \mathbf{II}(Z,X)\langle D_Y N, W\rangle \\ &= -\mathbf{II}(Z,Y)\mathbf{II}(X,W) + \mathbf{II}(Z,X)\mathbf{II}(Y,W) \quad . \end{aligned}$$

Now, if we fix v_1, v_2 a basis of T_pS and evaluate the last equation by setting $X = v_1, Y = v_2, Z = v_2, W = v_1$ and $h_{ij} = \mathbf{II}(v_i,v_j)$ we get

$$\begin{aligned} \langle R(v_1,v_2)v_2,v_1\rangle &= -\mathbf{II}(v_2,v_2)\mathbf{II}(v_1,v_1) + \mathbf{II}(v_2,v_1)\mathbf{II}(v_2,v_1) \\ &= -\left(\mathbf{II}(v_2,v_2)\mathbf{II}(v_1,v_1) - \mathbf{II}(v_1,v_2)^2\right) \\ &= -\det(h_{ij}) \quad . \end{aligned} \tag{7}$$

{eq:gc5}

Now, by definition of sectional curvature and setting $g_{ij} = \mathbf{I}(v_i, v_j)$,

$$\langle R(v_1, v_2)v_2, v_3\rangle = K(v_1, v_2)\det(g_{ij}) \quad . \tag{8}$$ {eq:gc6}

Here $K(v_1, v_2)$ is the sectional curvature of the plane spanned by v_1, v_2 which is simply the gaussian curvature K since we're in dimension 2.
So by putting together (7) and (8),

$$K = -\det_{\mathbf{I}}(\mathbf{II}) = -\frac{\det(h_{ij})}{\det(g_{ij})} \quad .$$

Which is the same as the Gauss equation for surfaces in Euclidean space if not for a negative sign.
Let's also focus on the other equation. We can define the exterior derivative of a tensor with respect to the connection, $d^{\nabla}T(X, Y) = \nabla_X T - \nabla_Y T$. This way, Codazzi's equation becomes

$$d^{\nabla}\mathbf{II} = 0.$$

So that the two equations get the following form:

$$\begin{cases} \det_{\mathbf{I}}(\mathbf{II}) = -K \\ d^{\nabla}\mathbf{II} = 0. \end{cases} \tag{9}$$ {eq:gc}

Bonnet's theorem
The Gauss-Codazzi equations play a fundamental role when the ambient manifold has constant sectional curvature. In this case, they are not only necessary compatibility conditions for the existence of an isometric immersion, but, on a simply connected surface, they are also sufficient.

Theorem C.1 (Bonnet). *Let $(g_{ij})_{ij}$ and $(h_{ij})_{ij}$ be differentiable functions defined on an open set $V \subset \mathbb{R}^2$ to $\mathbb{R}$ such that $g = (g_{ij})_{ij}$ is is a symmetric positive definite matrix and $h = (h_{ij})_{ij}$ is symmetric. Suppose that they satisfy the equation* (9).
Then for every $q \in V$ there exists a neighborhood $U \subset V$ of q and a diffeomorphism $\mathbf{x} : U \to \mathbf{x}(U) \subset \mathbb{M}^{2,1}$ such that the regular surface $\mathbf{x}(U)$ has g and h as first and second fundamental forms respectively.
Furthermore, if U is connected and if

$$\overline{\mathbf{x}} : U \to \overline{\mathbf{x}}(U) \subset \mathbb{M}^{2,1}$$

is another diffeomorphism satisfying the same conditions, then there exist an isometry $T : \mathbb{M}^{2,1} \to \mathbb{M}^{2,1}$ such that $\overline{\mathbf{x}} = T \circ \mathbf{x}$.

We redirect to [2] for a proof of the Euclidean case, whose only difference is the sign of Gauss' equation.

Mean curvature
We conclude this section by briefly recalling the definition of mean curvature, which does not differ from the usual Euclidean setting. As we have already seen, we can associate to the bilinear form $\mathbf{II}$ a linear operator $S_p : T_pS \to T_pS$ such that

$$\mathbf{II}(v, w) = \langle S_p(v), w\rangle \quad .$$

We define the mean curvature of S at p as

$$H(p) = \frac{1}{2}\operatorname{tr}(S_p) \quad .$$

If we again fix a basis v_1, v_2, denote with $(g_{ij})_{ij}$ and $(h_{ij})_{ij}$ the coefficients of the first and second fundamental forms, and express S_p as $S_p(v_i) = S_i{}^k v_k$, we get

$$h_{ij} = \mathbf{II}(v_i, v_j) = \langle S_p(v_i), v_j \rangle = S_i{}^k g_{kj} \quad .$$

If (g^{ij}) is the inverse matrix of (g_{ij}), it follows that

$$S_i{}^k = h_{ij} g^{jk},$$

and therefore

$$\operatorname{tr}(S_p) = S_i{}^i = h_{ij} g^{ji} = g^{ij} h_{ij} \quad .$$

So that we have

$$H = \frac{1}{2}\operatorname{tr}(S_p) = \frac{1}{2} S_i{}^i = \frac{1}{2} g^{ij} h_{ij} = \frac{1}{2}\operatorname{tr}_{\mathbf{I}}(\mathbf{II}) \quad .$$

Now, since

$$(g^{ij}) = \frac{1}{\det(g_{ij})} \begin{pmatrix} g_{22} & -g_{12} \\ -g_{12} & g_{11} \end{pmatrix} \quad ,$$

we obtain

$$2H = g^{ij} h_{ij} = \frac{h_{22} g_{11} - 2h_{12} g_{12} + h_{11} g_{22}}{\det(g_{ij})} \quad .$$

As we already mentioned, there is no formal difference with surfaces immersed in $\mathbb{R}^3$.

D Riemann surfaces and harmonic maps

We here introduce some notation and some fundamental concepts of Riemann surfaces. The most crucial part will be the fact that each metric on a surface is locally conformal to the Euclidean, the relationship between a Riemannian metric and a complex structure, the Uniformization theorem and the definition of the Hopf differential.

D.1 Preliminaries on Riemann surfaces

Let us fix S a surface (a topological manifold of dimension 2).

Definition D.1. A complex atlas $\mathcal{C}$ is a collection of charts

$$\mathcal{C} = \{z_\alpha : U_\alpha \to \mathbb{C}\}$$

that covers the whole surface S and whose transition maps

$$z_\beta \circ z_\alpha^{-1} : \; z_\alpha\big(U_\alpha \cap U_\beta\big) \to z_\beta\big(U_\alpha \cap U_\beta\big)$$

are holomorphic as maps between open subsets of $\mathbb{C}$. We call $\mathscr{C}$ a complex or conformal structure if it is an atlas maximal under the subset relation. Finally we call the pair $(S, \mathscr{C})$ (just S for short) a Riemann surface.

Definition D.2. A continuous map $h : S_1 \to S_2$ between Riemann surfaces is said to be holomorphic if, in local coordinates $\{U_\alpha, z_\alpha\}$ on S_1 and $\{U'_\beta, z'_\beta\}$ on S_2,all the maps $z'_\beta \circ h \circ z_\alpha^{-1}$ are holomorphic wherever they are defined. We call f a biholomorphism if it is holomorphic, bijective and its inverse f^{-1} is holomoprhic.
We say that the two Riemann surfaces S_1 and S_2 are isomorphic if there exists a biholomoprhism between the two.

Definition D.3. We say that two metrics g and g' on a surface S are conformal to each other if there exists a smooth function $u : S \to \mathbb{R}$ such that

$$g' = e^{2u} g \quad .$$

Equivalently, g' is at each point a positive multiple of g by a smooth function. The function e^{2u} is called the conformal factor.

Conformal structures on 2-surfaces
Here we state the results that lead to the fact that every Riemannian surface admits a complex structure.

Theorem D.1. *Given g a Riemannian metric on S differentiable surface, and a point $p \in S$, there is a chart near p, $\varphi : U \to \mathbb{R}^2$ such that g takes the form and a smooth positive function λ such that*

$$g = \lambda^2(dx^2 + dy^2) = e^{2v}(dx^2 + dy^2),$$

where (x, y) are the coordinates defined by φ.

Definition D.4. The chart (φ, U) in the theorem is called conformal, while the coordinates (x, y) on U are called isothermal coordinates around p.

Given a conformal chart we may introduce the change of coordinates

$$\begin{cases} z = x + iy \\ \bar{z} = x - iy \end{cases} \quad \text{or equivalently} \quad \begin{cases} x = \frac{1}{2}(z + \bar{z}) \\ y = \frac{i}{2}(z - \bar{z}) \end{cases} \quad .$$

The associated coordinated fields and forms can be expressed as

$$\begin{cases} \dfrac{\partial}{\partial z} = \dfrac{1}{2}(\dfrac{\partial}{\partial x} - i\dfrac{\partial}{\partial y}) \\ \dfrac{\partial}{\partial \bar{z}} = \dfrac{1}{2}(\dfrac{\partial}{\partial x} + i\dfrac{\partial}{\partial y}) \end{cases} \quad \text{and} \quad \begin{cases} dz = dx + idy \\ d\bar{z} = dx - idy \end{cases} \quad .$$

This lets us write the Euclidean metric as

$$\begin{aligned} dx^2 + dy^2 &= dx \otimes dx + dy \otimes dy \\ &= \frac{1}{4} d(z + \bar{z}) \otimes d(z + \bar{z}) - \frac{1}{4} d(z - \bar{z}) \otimes d(z - \bar{z}) \\ &= \frac{1}{2}(dz \otimes d\bar{z} + d\bar{z} \otimes dz) \\ &= \Re(dz \otimes d\bar{z}) \\ &= dz d\bar{z} \\ &= |dz|^2 \quad , \end{aligned}$$

where the last two equalities are definitions.
So we can express the metric g as

$$g = e^{2v}|dz|^2.$$

The change of coordinates $(x, y) \mapsto z = x + iy$ yields charts with values in $\mathbb{C}$. However, this is not yet enough to conclude that S carries a complex structure, since one still has to prove that the corresponding transition maps are holomorphic. But this can be shown with a little more work:

Proposition 3. *Given (φ, U) a conformal chart, denote with $(\tilde{\varphi}, U)$ the respective complex chart. Then*

$$\{(\tilde{\varphi}, U) \mid (\varphi, U) \textit{conformal charts on } (S, g)\}$$

constitutes a complex structure on S.

Now, let's recall some operators that take a simple form when read through a conformal chart, for which the metric has the form $e^{2u}|dz|^2$ First, the Laplacian with respect to the metric g is

$$\Delta_g = \frac{1}{\lambda^2}(\frac{\partial}{\partial x^2} + \frac{\partial}{\partial y^2}) = e^{-2u}(\frac{\partial}{\partial x^2} + \frac{\partial}{\partial y^2}) = e^{-2u}\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}}.$$

While the Gauss curvature of the surface takes the form

$$K_g = -\Delta_g \log(\lambda) = -2\Delta u.$$

Let us finally observe that,

$$\begin{aligned} dz^2 &= dz \otimes dz \\ &= d(x + iy) \otimes d(x + iy) \\ &= dx \otimes dx + i\,(dx \otimes dy + dy \otimes dx) - dy \otimes dy \\ &=: dx^2 - dy^2 + 2idxdy \quad . \end{aligned}$$

Integration through a conformal chart
Given a curve γ in S, its length is defined by

$$\ell(\gamma) = \int_\gamma \lambda(z)|dz| \ .$$

On the other hand the area form can be expressed as follows

$$dx \wedge dy = -\frac{i}{2}dz \wedge d\bar{z} \ ,$$

so that the area of a measurable set becomes

$$\begin{aligned} \text{Area}(B) &:= \int_B \lambda^2(z)\frac{i}{2}dz \wedge d\bar{z} \\ &= \int_B \lambda^2(z)\frac{i}{2}dzd\bar{z} \quad . \end{aligned}$$

D.2 Uniformization

An important property of Riemann surfaces is that they can be classified in just three categories, depending on their universal covering. This result is (a direct consequence of) the uniformization theorem. In what follows we denote by $\widehat{\mathbb{C}}$ the 2-sphere, with the metric and conformal structure induced by $\mathbb{R}^3$ or equivalently $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$.

Theorem D.2 (Uniformization Theorem)**.** *Let Σ be a simply connected Riemann surface. Then Σ is isomorphic to exactly one of the following three surfaces:*

1. *The Riemann sphere $\widehat{\mathbb{C}}$.*
2. *The complex plane $\mathbb{C}$.*
3. *The Poincaré disk $\mathbb{D}$.*

Every Riemann surface Σ admits a universal covering $\tilde{\Sigma}$, which inherits the complex structure of its base space. Since the universal covering $\tilde{\Sigma}$ of a surface Σ is simply connected, we can apply the uniformization theorem to it, making Σ fall in one of the following categories. We say that Σ is

- elliptic if $\widetilde{\Sigma} \cong \widehat{\mathbb{C}}$,
- parabolic if $\widetilde{\Sigma} \cong \mathbb{C}$,
- hyperbolic if $\widetilde{\Sigma} \cong \mathbb{D}$.

In our case the surface we consider is simply connected and noncompact (for instance because it's the graph of an entire function on $\mathbb{R}^2$) so we can exclude the elliptic case.

Corollary D.1. *If Σ is a simply connected noncompact Riemann surface, then Σ is conformally equivalent either to $\mathbb{C}$ or to $\mathbb{D}$.*

D.3 Harmonic maps between Riemann surfaces

We here recall the definition of harmonic map between Riemann surfaces and some important facts related to them. Let (Σ_1, g) and (Σ_2, h) two Riemann surfaces. Suppose to have fixed a conformal parameter z on Σ_1 and u on Σ_2 so that the we can write the two metrics as

$$\lambda^2(z)|dz|^2 \qquad \text{and} \qquad \varrho^2(u)|du|^2 \quad .$$

Let $f : \Sigma_1 \to \Sigma_2$ be a smooth map. For brevity, we use the following notation $f_z = \frac{\partial f}{\partial z}, f_{\bar{z}} = \frac{\partial f}{\partial \bar{z}}$. Also, it's immediate to see that $\overline{f_z} = \overline{f}_{\overline{z}}$.

Definition D.5. We define the energy density of a map as

$$\begin{aligned} e(f)(z) &= \frac{\varrho^2(f(z))}{\lambda^2(z)} \left(|f_z|^2 + |f_{\overline{z}}|^2\right) \\ &= \frac{\varrho^2(f(z))}{\lambda^2(z)} \left(f_z \overline{f}_{\overline{z}} + \overline{f}_z f_{\overline{z}}\right) \quad . \end{aligned}$$

The integral of this quantity is the energy of the map

$$\begin{aligned} E(f) &:= \int_{\Sigma} \lambda^2(z) e(f(z)) i dz d\overline{z} \\ &= \int_{\Sigma_1} \varrho^2(f)\big(f_z \overline{f}_{\overline{z}} + \overline{f}_z f_{\overline{z}}\big)\frac{i}{2} dz d\bar{z} \quad . \end{aligned}$$

This definition is given in local coordinates but it's actually well posed, as proven by the following lemma whose proof can be found in [7]

Lemma 1. *$E(u)$ is independent of the choice of conformal parameters on Σ_1 and Σ_2.*

This means we have defined a functional E on the C^2 functions between Σ_1 and Σ_2. The critical points of such a functional are what we call the Harmonic maps between Σ_1 and Σ_2. This functional has an Euler-Lagrange equation associated to it, which sufficies as our definition of harmonic map.

Definition D.6. A map $f \in C^2(\Sigma_1, \Sigma_2)$ is called harmonic if

$$f_{z\overline{z}} + \frac{2\varrho_u}{\varrho} f_z f_{\overline{z}} = 0 \ .$$

An important observation is that the local harmonic map equation does not involve the conformal factor λ of the metric on the domain. This reflects the fact that, in dimension two, harmonicity depends only on the conformal class of the metric on the source surface.

D.4 Energy densities and the Hopf differential

Energy densities and Jacobian

A related concept to the energy of the map f are the Holomorphic and anti-holomorphic energy densities.

Definition D.7. We define the holomorphic and antiholomorphic energy density of a smooth map $f : \Sigma_1 \to \Sigma_2$ respectively as

$$H(f) := \frac{\varrho^2(f)}{\lambda^2}|f_z|^2, \qquad \text{and} \qquad L(f) := \frac{\varrho^2(f)}{\lambda^2}|f_{\bar{z}}|^2 \quad .$$

It is immediate to note that

$$e(f) = H(f) + L(f) \ .$$

The Jacobian of the map is by definition

$$J(f) = \frac{\varrho^2(f)}{\lambda^2}(|f_z|^2 - |f_{\overline{z}}|^2) \quad .$$

So that we also have

$$J(f) = H(f) - L(f) \ . \quad . \tag{10}$$ {eq:defjacob}

Quadratic differentials
To each map between Riemann surfaces we can associate a quantity that carries information about how far the map is from being conformal.

Definition D.8. Given Σ a Riemann surface and a local coordinate z, we say that the 2-form $\varphi(z)dz^2$ is a holomorphic quadratic differential if φ is holomorphic.

It is clear that if we ha a change of coordiante $z \mapsto w = \gamma(z)$, then we can express the differentials as $dw = \gamma'(z)dz$. This means that the tensor product dz^2 will transform as

$$dw^2 = \gamma'(z)^2 dw^2.$$

To each map between Riemann surfaces, we can associate a quadratic differential, which satisfies some properties if it is harmonic

{lemma:exholdiff}

Lemma 2. *Let $f : \Sigma_1 \to \Sigma_2$ a harmonic map between Riemann surfaces, with Σ_1 being endowed with a metric of nonpositive curvature. Let z be a conformal parameter on Σ_1. Then if w is a local representation of f,*

$$\varrho^2(w) w_z w_{\overline{z}} dz^2$$

is a holomorphic quadratic differential on Σ_1. Furthermore we have that $\varrho^2(w) w_z w_{\overline{z}} \equiv 0$ if and only if f is holomorphic or antiholomorphic.

We can actually give a decomposition of any symmetric 0-2 tensor field

Lemma 3. *Let ω a symmetric 0-2 tensor field on (Σ, g) with values in $\mathbb{C}$. Then Ω can be decomposed uniquely as*

$$\omega = \omega^{(2,0)} + \omega^{(1,1)} + \omega^{(0,2)},$$

where $\omega^{(2,0)}$ and $\overline{\omega^{(0,2)}}$ are holomorphic quadratic differentials and $\omega^{(1,1)}$ is a multiple of the metric g.

We can then apply this last lemma to lemma to the differential we found in lemma 2, obtaining

Lemma 4. *Let $f : (\Sigma_1, g) \to (\Sigma_2, h)$ a map two Riemann surfaces with Σ_1 being hyperbolic. Let q be the quadratic differential associated to f as in lemma 2. Then*

$$q = (f^*(h))^{(2,0)} \ .$$

Moreover we have that

$$f^*(h) = q + \frac{e}{2} g + \bar{q} \ .$$

This motivates the definition of the Hopf differential

Definition D.9. Given $f : (\sigma_1, g) \to (\Sigma_2, h)$ between two Riemann surfaces, we define the Hopf differential as the quadratic differential

$$\mathrm{Hopf}(f) = (f^*(h))^{(2,0)} \ .$$

We can calculate, pointwise, a norm for a quadratic differential. Fix $(e_i)_i$ a local orthonormal frame on (Σ_1, g). Let's define the g norm of the quadratic differential q as

$$\|q\|_g = \frac{1}{4}\sum_{i,j} |q(e_i, e_j)|^2 \ .$$

It's possible to rewrite this quantity in terms of the energies of the map f. If the metric is $\lambda^2|dz|^2$, two orthonromal vectors are $e_1 = \frac{1}{\lambda}\frac{\partial}{\partial x}$ and $e_2 = \frac{1}{\lambda}\frac{\partial}{\partial y}$, hence

$$dz^2(e_1, e_1) = \frac{1}{\lambda^2},\ dz^2(e_1, e_2) = i\frac{1}{\lambda^2},\ dz^2(e_2, e_2) = -\frac{1}{\lambda^2} \ .$$

So that we have

$$\begin{aligned}
\|q\|_g^2 &= \frac{1}{4}\sum_{ij} |q(e_i, e_j)|^2 \\
&= \frac{1}{4}\left(|q(e_1, e_1)|^2 + 2|q(e_1, e_2)|^2 + |q(e_2, e_2)|^2\right) \\
&= \frac{1}{4}|q(z)|^2\left(|dz(e_1, e_1)|^2 + 2|q(e_1, e_2)|^2 + |q(e_2, e_2)|^2\right) \\
&= \frac{|q(z)|^2}{\lambda^4(z)} \ .
\end{aligned}$$

If we apply calculate the norm of the Hopf differential of a smooth $f : \Sigma_1 \to \Sigma_2$, with local representation $w = w(z)$,

$$\|\operatorname{Hopf}(f)\|_g^2(z) = \frac{\varrho^4(w)}{\lambda^4}|w_z|^2|w_{\overline{z}}|^2 \ .$$

By recognizing the terms in this last equation we have immediately the following lemma.

{lemma:qnorm}

Lemma 5. *Let $q = \operatorname{Hopf}(f)$. Then*

$$\|q\|^2 = HL \ ,$$

where H and L are respectively the holomorphic and anti-holomorphic energy of f.

An important result we will need is the following

Theorem D.3 (Bochner identities)**.** *At the points where $H \neq 0$, if K_1 and K_2 are the Gaussian curvatures of Σ_1 and Σ_2 respectively, we have*

$$\frac{1}{2}\log H = K_1 - K_2 J \ . \tag{11}$$

{eq:bochner}

And similarly, where $L \neq 0$,

$$\frac{1}{2}\log L = K_1 + K_2 J \ .$$

E CMC surfaces and harmonic maps

We now introduce the class of surfaces that will be studied throughout this section.

Definition E.1. A constant mean curvature cut is a complete entire spacelike surface $S \subset \mathbb{M}^{2,1}$ with constant mean curvature $H > 0$.

This is essentially the same setting we exposed in section B if not for the additional condition of having a constant mean curvature. The first step for the desired classification is the following proposition, which we essentially proved in section B if not for the last result, which can be found in [4]

Proposition 4. *Let S be a complete spacelike surface in $\mathbb{M}^{2,1}$ with constant mean curvature H. Then S can be represented as the graph of an entire function $x^3 = f(x^1, x^2)$. Moreover, f is convex, and therefore the Gaussian curvature of S satisfies $K \leq 0$.*

An immediate consequence of being the graph of a convex function is that the second fundamental form of the surface is positive definite, from which we get that the mean curvature must be positive.

We observe now that we may suppose the mean curvature of S to be 1. In fact, if $\iota : S \to \mathbb{M}^{2,1}$ is an immersion with constant mean curvature $H = k > 0$, then we may consider the dilated immersion

$$\tilde{\iota} = \mu\iota,$$

which has constant mean curvature $\tilde{H} = \mu^{-1}k$. Choosing $\mu = k$, we obtain a surface with constant mean curvature 1.

Therefore, there exists a bijection between complete space-like surfaces with constant mean curvature k and complete space-like surfaces with constant mean curvature 1. So a classification of the latter immediately yields a classification of the former. We may then assume from now on that the surfaces have constant mean curvature 1.

The following is the reason for our focus on Harmonic maps, whose proof can be found in [8] and [5]

Lemma 6. *The Gauss map of a spacelike surface in $\mathbb{M}^{2,1}$*

$$u : S \to \mathbb{D}$$

is harmonic if and only if the mean curvature of S is constant.

Now, we assumed S to be hyperbolic, so the uniformization theorem provides a conformal diffeomorphism

$$\Lambda : \mathbb{D} \to S.$$

By composing the Gauss map with Λ, we may therefore read the Gauss map as a map

$$u : \mathbb{D} \to \mathbb{D}.$$

Since harmonicity is invariant under conformal changes of the metric on the domain, this new expression of u is still harmonic.

Now, the map u lets us pull back the metric of the hyperbolic disk.

{prop:indmetr}

Proposition 5. *Suppose that S has constant mean curvature 1. Then the pulled-back induced metric $g = \Lambda^*(\mathbf{I})$ of S and the respective Gaussian curvature are given by*

$$g = \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2, \qquad K = \frac{|u_{\overline{z}}|^2}{|u_z|^2} - 1 \quad .$$

Proof. We redirect the reader to [1]. □

We now need a claim we will prove later in the section.

{cl:jhupos}

Claim 1. The Gauss map u satisfies

$$J(u) \geq 0 \qquad \text{and} \qquad H(u) > 0 \ .$$

Being u harmonic, its Hopf differential is a holomorphic quadratic differential. So, if we denote by

$$\mathrm{Hopf}(f)(z) = \varphi(z)dz^2$$

then $\varphi(z) = \varrho^2(u(z))u_z\overline{u}_{\overline{z}}$ is holomorphic. For brevity we set $\Phi := \mathrm{Hopf}(f)$

Let's now observe that, from lemma 5 we have

$$L(u) = \|\Phi\|^2 H^{-1}(u).$$

So, from (10) we get

$$\begin{aligned} J(u) &= H(u) - L(u) \\ &= H(u) - \|\Phi\|^2 H^{-1}(u) \quad . \end{aligned} \tag{12}$$

{eq:cmcclass1}

Now, let's define w as

$$w = \frac{1}{2}\log H \ ,$$

which is well defined because $H > 0$ by our claim.
We can then write $H = e^{2w}$ and therefore substituting in (12),

$$J(u) = e^{2w} - \|\Phi\|^2 e^{-2w} \ .$$

Now, let's consider the Bochner identity (11), applied to u (which is valid because $H > 0$),

$$\Delta_{\Sigma_1}\log H = -2K_2 J(u) + 2K_1 \quad .$$

By substituting for w and J, we have

$$2\Delta_{\Sigma_1} w = -2K_2(e^{2w} - \|\Phi\|^2 e^{-2w}) + 2K_1 \quad .$$

We recall that $u : \mathbb{D} \to \mathbb{D}$, so in the equation we actually have $\Sigma_1 = \Sigma_2$ and so $K_1 \equiv K_2 \equiv -1$. Hence we get

$$\Delta_{\Sigma_1} w = e^{2w} - \|\Phi\|^2 e^{-2w} - 1 \quad .$$

Let us prove the claim

Proof of claim 1. First, by proposition 5 we have that the metric $g := \Lambda^*(\mathbf{I}_S)$ is expressed by

$$g = \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2 \ .$$

Then we recall that the metric on $\mathbb{D}$ at a point v is $ds_p^2|_v := \frac{4}{(1-|v|^2)^2}|dz|^2$. So we have

$$\begin{aligned} g &= \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2 \\ &= ds_p^2|_u|u_z|^2 \\ &= ds_p^2|_u H(u) \ \ . \end{aligned}$$

We know that both g and ds_p^2 are Riemannian metrics, so that we obtain $H(u) > 0$.

Now, from the second equation of proposition 5 we havee

$$K = \frac{|u_{\overline{z}}|^2}{|u_z|^2} - 1$$

so that

$$\begin{aligned} |u_z|^2 K &= |u_{\overline{z}}|^2 - |u_z|^2 \\ &= -J(u) \ \ . \end{aligned}$$

Then

$$J(u) = -|u_z|^2 K \geq 0 \ ,$$

since $K \leq 0$ □

From the proof we can also observe that the metric induced on S can be expressed as $g = H ds_p^2 = e^{2w} ds_p^2$.

F Classification of hyperbolic CMC cuts

G Uniqueness for the PDE

So far we have reduced our classification problem to proving existence and uniqueness for the following differential problem.

Given q a smooth function on the Poincaré disk $\mathbb{D}$, find a smooth function

$$w : \mathbb{D} \to \mathbb{R}$$

such that

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1 \ , \tag{13}$$ {eq:box}

where Δ_p is the Laplacian with respect to the metric ds_p^2 and such that

$$e^{2w} ds_p^2$$

is a complete metric on D, with

$$e^{2w} - q^2 e^{-2w} > 0 \ . \tag{14}$$ {eq:box2}

on $\mathbb{D}$.

rimuoverò questo box una volta scritte le sezioni precedenti

We will need the following generalized maximum principle, due to Cheng and Yau and found in the article [3].

G.1 A generalized maximum principle

Let M be a complete Riemannian manifold of dimension n, and fix a point $p \in M$. Assume moreover that M is simply connected and has nonpositive sectional curvature. This assumption is stronger than what is strictly needed in this section, but greatly simplifies the discussion and is satisfied in the case of a complete space-like surface of $\mathbb{M}^{2,1}$. These hypotheses allow us to apply the Cartan-Hadamard theorem, so that the exponential map $\exp_p$ is a diffeomorphism. In particular, this tells us that for every $x \in M$ there exists a unique geodesic joining p to x, and that $d(\cdot, \cdot)$ is smooth on $M \setminus \{x = y\}$.

We need now to define an important quantity we will refer to multiple times later. Let us fix a point $x \in M$ and let σ be the unique unit speed geodesic that joins x to p. In the following expression we also denote with l the length of σ, or in other terms $l = d(x, p)$. Let $\sigma'(t)$ be the velocity vector of σ, and denote with $\mathrm{Ric}(\sigma'(t))$ the Ricci curvature along σ. We then define

$$K(x) = \min_{k \in [0,l]} \left[\frac{n-1}{l-k} - \frac{1}{(l-k)^2} \int_k^l (t-k)^2 \, \mathrm{Ric}(\sigma'(t)) dt \right] \ . \tag{15}$$ {eq:intromaxprin1}

We now claim that, if $r(x) \geq 1$ and the Ricci curvature is bounded from below, then $K(x)$ is bounded from above by a constant independent of x. Indeed, if we assume that for every unit vector v, $\mathrm{Ric}(v) \geq \mathscr{S}$, then

$$\frac{1}{(l-k)^2} \int_k^l (t-k)^2 \, \mathrm{Ric}(\sigma'(t)) \, dt \geq \frac{1}{(l-k)^2} \int_k^l (t-k)^2 \mathscr{S} \, dt = \frac{(l-k)}{3} \mathscr{S} \ .$$

Therefore in the argument of min in (15) we have

$$\frac{n-1}{l-k} - \frac{1}{(l-k)^2} \int_k^l (t-k)^2 \, \mathrm{Ric}(\sigma'(t)) \, dt \leq \frac{n-1}{l-k} - \frac{(l-k)}{3} \mathscr{S} \ .$$

Since $r(x) = l \geq 1$, if we choose $k = l - 1$ we get

$$K(x) \leq n - 1 - \frac{\mathscr{S}}{3} \quad .$$

Which proves our claim.

The next lemma, due to Yau, provides the estimate for the distance function that will be used in the proof of Theorem G.1.

{lm:distest}

Lemma 7. *With the notation introduced above, for every $x \in M \setminus \{p\}$ such that $r(x) \geq 1$ we have*

$$\Delta r(x) \leq K(x) \ .$$

Proof. We redirect the reader to [10] for a proof. □

We are ready to state and prove the maximum principle.

{th:maxprin}

Theorem G.1. *Let M be a complete, simply connected Riemannian manifold of nonpositive sectional curvature, with Ricci curvature bounded from below. Let $f \in C^2(M)$ be bounded from above. Fix $p \in M$, and let $r(x) = d(p, x)$. Then there exist a constant $C > 0$ and a sequence $(x_k)_k \subset M$ such that*

$$f(x_k) \to \sup_M f, \qquad |\nabla f|(x_k) \leq \frac{C}{k}, \qquad \Delta f(x_k) \leq \frac{C}{k}.$$

In particular,

$$\limsup_{k\to\infty} \Delta f(x_k) \leq 0.$$

Proof. Let's define the auxiliary functions $u(x) = (f(x) - f(p) + 1)$ and $L(x) = \log(r^2(x) + 2)$. Then let's fix a positive integer k and define the function g as

$$g(x) = \frac{f(x) - f(p) + 1}{(\log(r^2(x) + 2)^{\frac{1}{k}}} = \frac{u(x)}{L(x)^{\frac{1}{k}}} \quad .$$

Let's observe that $g(p) = \frac{1}{\log(2)^{\frac{1}{k}}} > 0$ and that $\lim_{r(x)\to\infty} g(x) = 0$. Since g is continuous, this means that it attains its maximum at a certain point x_k.

First, let's show that we can assume that $r(x_k) \to \infty$. If that's not the case, up to taking a subsequence, $r(x_k)$ is bounded. Then we can find a closed geodesic ball $\overline{B}(p; \varrho)$ with $\varrho > 0$ that contains x_k for every k sufficiently big. Since M is complete, the Hopf Rinow theorem implies that $\overline{B}(p; \varrho)$ is compact. Therefore, after passing to a subsequence, we may assume that

$$x_k \to x_0 \quad .$$

But we know that $f(x_k) \to \sup f$ and that f is continuous, which implies that x_0 is a maximum point.
This entails that

$$\nabla f(x_0) = 0 \qquad \text{and} \qquad \Delta f(x_0) \leq 0 \quad .$$

Which verifies trivially the thesis by setting $x_k \equiv x_0$. Thus we assume, from now on, that $r(x_k) \to \infty$.

Let us prove first that

$$\limsup_{k\to\infty} f(x_k) = \sup_M f \quad . \tag{16}$$ {eq:boundprin13}

Assume it by contradiction to be false. Then by definition of sup and lim sup there exists a $\delta > 0$ and a $\bar{x} \in M$ such that for k sufficiently big,

$$f(x_k) < f(\bar{x}) - \delta \ , \qquad \text{equivalently,} \qquad f(x_k) + \delta < f(\bar{x}) \quad . \tag{17}$$ {eq:boundprin12}

Since we assumed that $r(x_k) \overset{k\to\infty}{\longrightarrow} \infty$, then we have, for k sufficiently large, $r(x_k) > r(\bar{x})$ and $f(\bar{x}) > f(x_k)$ so that

$$g(\bar{x}) = \frac{f(\bar{x}) - f(p) + 1}{(\log(r^2(\bar{x}) + 2)^{\frac{1}{k}}} > \frac{f(x_k) - f(p) + 1}{(\log(r^2(x_k) + 2)^{\frac{1}{k}}} = g(x_k) \quad .$$

Which contradicts the maximality of x_k. It is obvious that, up to taking a subsequence, we can assume $f(x_k) \to \sup_M f$.

Let's now focus on the two estimates. As we already mentioned, the Cartan-Hadamard M theorem guarantees that the function $d(p,x)^2 = r(x)^2$ is differentiable in x_k.
We can easily calculate the gradient of g,

$$\nabla g(x) = \frac{\nabla f(x)}{\log(r(x)^2 + 2)^{1/k}} - \frac{2(f(x) - f(p) + 1)r(x)}{k(r(x)^2 + 2)\log(r(x)^2 + 2)^{1+1/k}} \nabla r(x) \quad .$$

Since x_k is a maximum point of g_k, we have $\nabla g(x_k) = 0$. Hence

$$\nabla f(x_k) = \frac{2(f(x_k) - f(p) + 1)\, r(x_k)}{k\,(r^2(x_k) + 2)\ \log(r^2(x_k) + 2)} \nabla r(x_k) \ , \tag{18}$$ {eq:boundprin9}

$$|\nabla f|(x_k) = \frac{2(f(x_k) - f(p) + 1)\, r(x_k)}{k\,(r^2(x_k) + 2)\ \log(r^2(x_k) + 2)} \ , \tag{19}$$ {eq:boundprin8}

where we used $|\nabla r(x_k)| = 1$.
We then observe, since f is bounded from above, also u is

$$u(x) = (f(x) - f(p) + 1) \le \sup_M f - f(p) + 1 =: A.$$

Also, it's not difficult to show that the real function satisfies

$$\frac{2x}{(x^2 + 2)\log(x^2 + 2)} \le 2 \ , \quad . \tag{20}$$ {eq:boundprin14}

so that by substituting in (19) we get the desired estimate

$$|\nabla f|(x_k) \le \frac{2A}{k} \quad .$$

Let's now focus on the estimate on the Laplacian. We can find the expression of the Laplacian of $g = u/L^{\frac{1}{k}}$. First we observe that

$$\begin{aligned}\Delta g &= \nabla \cdot \nabla \left(\frac{u}{L^{\frac{1}{k}}} \right) \\ &= \nabla \cdot \left(\nabla u L^{-\frac{1}{k}} + u \nabla (L^{-\frac{1}{k}}) \right) \\ &= \frac{\Delta u}{L^{\frac{1}{k}}} + 2 \left\langle \nabla u, \nabla (L^{-\frac{1}{k}}) \right\rangle + u \Delta (L^{-\frac{1}{k}}) \quad , \end{aligned} \tag{21}$$

{eq:boundprindeltag1}

then we calculate the involved quantities. First,

$$\nabla u = \nabla f, \qquad \nabla L = \frac{2r\nabla r}{r^2+2}, \qquad \nabla \left(\frac{1}{r^2+2} \right) = -\frac{2r\nabla r}{(r^2+2)^2},$$

$$\nabla \left(L^{-\frac{1}{k}} \right) = -\frac{1}{k} \frac{\nabla L}{L^{\frac{1}{k}+1}} = -\frac{1}{k} \frac{2r\nabla r}{(r^2+2)L^{\frac{1}{k}+1}},$$

$$\nabla \left(L^{-\frac{1}{k}-1} \right) = \left(-\frac{1}{k} - 1 \right) \frac{\nabla L}{L^{\frac{1}{k}+2}} = -\frac{k+1}{k} \frac{2r\nabla r}{(r^2+2)L^{\frac{1}{k}+2}},$$

$$\Delta L = \nabla \cdot \left(\frac{2r\nabla r}{r^2+2} \right) = \frac{2|\nabla r|^2}{r^2+2} + \frac{2r\Delta r}{r^2+2} - \frac{4r^2|\nabla r|^2}{(r^2+2)^2} \quad ,$$

and then

$$\begin{aligned}\Delta \left(L^{-\frac{1}{k}} \right) &= \nabla \cdot \nabla L^{-\frac{1}{k}} \\ &= -\frac{1}{k} \nabla \cdot \left(\frac{\nabla L}{L^{\frac{1}{k}+1}} \right) \\ &= -\frac{1}{k} \frac{\Delta L}{L^{\frac{1}{k}+1}} - \frac{1}{k} \left\langle \nabla L, \nabla \left(L^{-\frac{1}{k}-1} \right) \right\rangle \\ &= -\frac{1}{k} \left(\frac{2|\nabla r|^2}{r^2+2} + \frac{2r\Delta r}{r^2+2} - \frac{4r^2|\nabla r|^2}{(r^2+2)^2} \right) \frac{1}{L^{\frac{1}{k}+1}} + \frac{k+1}{k^2} \left\langle \frac{2r\nabla r}{r^2+2}, \frac{2r\nabla r}{(r^2+2)L^{\frac{1}{k}+2}} \right\rangle \\ &= -\frac{2}{k} \frac{1}{(r^2+2)L^{\frac{1}{k}+1}} - \frac{2}{k} \frac{r\Delta r}{(r^2+2)L^{\frac{1}{k}+1}} + \frac{4}{k} \frac{r^2}{(r^2+2)^2 L^{\frac{1}{k}+1}} + \frac{4(k+1)}{k^2} \frac{r^2}{(r^2+2)^2 L^{\frac{1}{k}+2}} \quad . \end{aligned}$$

Hence, by substituting in (21) and simplifying, we obtain

$$\begin{aligned}\Delta g = \frac{\Delta f}{L^{1/k}} &- \frac{4r}{k(r^2+2)L^{1+1/k}} \langle \nabla f, \nabla r \rangle - \frac{2u}{k(r^2+2)L^{1+1/k}} \\ &- \frac{2ur\,\Delta r}{k(r^2+2)L^{1+1/k}} + \frac{4ur^2}{k(r^2+2)^2 L^{1+1/k}} + \frac{4(k+1)ur^2}{k^2(r^2+2)^2 L^{2+1/k}} \quad . \end{aligned}$$

Since x_k is a maximum point of g, we have $\Delta g(x_k) \leq 0$. Using the expression for Δg, the identity $\nabla g(x_k) = 0$, and

$$\langle \nabla f, \nabla r \rangle (x_k) = \frac{2u(x_k)r(x_k)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \quad ,$$

which we have derived from (18), we obtain

$$\begin{aligned}\Delta f(x_k) \leq \frac{2u(x_k)\big(1 + r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} &- \frac{4u(x_k)r^2(x_k)}{k(r^2(x_k)+2)^2 \log(r^2(x_k)+2)} \\ &- \frac{4(k-1)u(x_k)r^2(x_k)}{k^2(r^2(x_k)+2)^2 \log^2(r^2(x_k)+2)} \quad . \end{aligned}$$

In particular, by discarding the last two nonpositive terms, we get

$$\Delta f(x_k) \leq \frac{2u(x_k)\big(1 + r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \ . \tag{22}$$ {eq:boundprin10}

First we find another inequality for a real function, namely

$$\frac{1}{(x^2+2)\log(x^2+2)} \leq 2 \ . \tag{23}$$ {eq:boundprin20}

, it follows from (22),

$$\begin{aligned}
\Delta f(x_k) &\leq \frac{2u(x_k)\big(1 + r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \\
[u \leq A] \qquad &\leq \frac{2A}{k}\left(\frac{1}{(r^2(x_k)+2)\log(r^2(x_k)+2)} + \frac{r(x_k)\Delta r(x_k)}{(r^2(x_k)+2)\log(r^2(x_k)+2)}\right) \\
[(20),\ (23)] \qquad &\leq \frac{2A}{k}\left(2 + 2\Delta r(x_k)\right) \ . \qquad (24)
\end{aligned}$$ {eq:boundprin11}

We now apply the estimate of Lemma 7. We can do this because under the assumption that $r(x_k) \to \infty$ we have $r(x_k) \geq 1$ for k sufficienly large, so that

$$\Delta r(x) \leq K(x) \ .$$

But, as we already observed, $K(x)$ is uniformly bounded from above when $r(x) \geq 1$ since the Ricci curvature is bounded from below.
So we conclude in (24) that there is a constant $\tilde{C}$ for which

$$\Delta f(x_k) \leq \frac{\tilde{C}}{k} \ .$$

□

A boundedness result
We will also use this theorem to obtain a boundedness result for a certain differential inequality. We first make a few preliminary observations, and then state the theorem at the end of the section.

Let M be a complete Riemannian manifold with Ricci curvature bounded from below, and let $u \in C^2(M)$ satisfy

$$\Delta u \geq f \ , \tag{25}$$ {eq:boundprin2}

where the assumptions on f will be specified later.
Now let $F \in C^2(\mathbb{R})$ be a positive function, whose precise form will also be given later, which we require to be strictly increasing or in other terms to have a strictly positive derivative. Since $F > 0$, the function $-1/F$ is bounded from above. We may therefore apply Theorem G.1 to $-1/F(u)$ and obtain a sequence $(x_k)_k$ such that

$$\lim_{k\to\infty} \frac{1}{F(u(x_k))} = \inf \frac{1}{F(u)} = -\sup\left(-\frac{1}{F(u)}\right) . \tag{26}$$ {eq:boundprin7}

Moreover, the corresponding gradient and Laplacian estimates hold:

$$\frac{|F'(u(x_k))|\,|\nabla u|(x_k)}{F(u(x_k))^2} \leq \frac{1}{k}, \tag{27}$$ {eq:boundprin3}

$$-\frac{F''(u(x_k))\,|\nabla u|^2(x_k)}{F(u(x_k))^2} - \frac{F'(u(x_k))\,\Delta u(x_k)}{F(u(x_k))^2} + \frac{2(F'(u(x_k)))^2\,|\nabla u|^2(x_k)}{F(u(x_k))^3} \geq -\frac{1}{k}\ . \tag{28}$$ {eq:boundprin16}

For simplicity, from now on we evaluate all quantities at x_k.
Then, multiplying (28) by $\dfrac{F'(u)^2}{F(u)^2|F''(u)|}$, we obtain

$$-\frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} - \frac{F'(u)^3}{F(u)^4}\frac{\Delta u}{|F''(u)|} + \frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \geq -\frac{F'(u)^2}{kF(u)^2|F''(u)|}\ . \tag{29}$$ {eq:boundprin1}

We now estimate the terms on the left-hand side of (29). First,

$$-\frac{F'(u)^3}{F(u)^4}\frac{\Delta u}{|F''(u)|} \leq -\frac{F'(u)^3}{F(u)^4}\frac{f(u)}{|F''(u)|}\ . \qquad \text{[by (25), } F'>0\text{]}$$

Moreover,

$$-\frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \leq \left|\frac{F'(u)^2|\nabla u|^2}{F(u)^4}\right| \leq \frac{1}{k^2}\ . \qquad \text{[by (27)]}$$

Also,

$$\begin{aligned}\frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} &\leq 2\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right|\left|\frac{F'(u)^2|\nabla u|^2}{F(u)^4}\right| \\ &\leq \frac{2}{k^2}\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right|\ . \qquad \text{[by (27)]}\end{aligned}$$

Therefore, from (29) we obtain

$$\frac{1}{k^2} - \frac{F'(u)^3 f(u)}{F(u)^4|F''(u)|} + \frac{2}{k^2}\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right| \geq -\frac{1}{k}\frac{F'(u)^2}{F(u)^2|F''(u)|}\ . \tag{30}$$ {eq:boundprin6}

Suppose now that F satisfies

$$\limsup_{t\to\infty}\frac{|F'(t)|^2}{F(t)\,|F''(t)|} < \infty\ , \tag{31}$$ {eq:boundprin4}

and

$$\liminf_{t\to\infty}\frac{F'(t)^3 f(t)}{F(t)^4\,|F''(t)|} > 0\ . \tag{32}$$ {eq:boundprin5}

We claim that this implies

$$\limsup_{k\to\infty} u(x_k) < \infty\ .$$

Indeed, suppose by contradiction that $\limsup_{k\to\infty} u(x_k) = \infty$. Passing to a subsequence if necessary we may assume that

$$u(x_k) \to \infty \quad .$$

Taking lim sup in (30), and using (31) together with the fact that later our choice of F will satisfy $F \geq 1$, we obtain

$$-\liminf_{k\to\infty} \frac{F'(u(x_k))^3 f(u(x_k))}{F(u(x_k))^4 |F''(u(x_k))|} \geq 0 \quad ,$$

that is,

$$\liminf_{k\to\infty} \frac{F'(u(x_k))^3 f(u(x_k))}{F(u(x_k))^4 |F''(u(x_k))|} \leq 0 \quad ,$$

which contradicts (32).

So we have proved so far that

$$\limsup_{k\to\infty} u(x_k) < \infty \quad .$$

Passing to a subsequence if necessary, we may therefore assume that

$$u(x_k) \to \ell$$

for some $\ell \in \mathbb{R}$.

Let us now show that u is bounded from above.
Since $F > 0$, from (26) we obtain

$$\lim_{k\to\infty} \frac{1}{F(u(x_k))} = \inf \frac{1}{F(u)} = \frac{1}{\sup_M F(u)} \quad ,$$

and therefore

$$F(u(x_k)) \to \sup F(u) \quad .$$

On the other hand, by continuity of F and the fact that $u(x_k) \to \ell$, we also have

$$F(u(x_k)) \to F(\ell) \quad .$$

Hence

$$F(\ell) = \sup F(u) \quad .$$

Since F is strictly increasing, for every $x \in M$ we have

$$F(u(x)) \leq F(\ell) \qquad \Longrightarrow \qquad u(x) \leq \ell \quad .$$

This means that ℓ is an upper bound for u, and hence u is bounded from above:

$$\sup u < \infty \quad .$$

We have therefore reduced the problem to finding a function F which is positive, strictly increasing, and satisfies (31) and (32). Let us search for such a function of the form

$$F(s) = \int_a^s \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} dt + 1 \quad , \tag{33}$$ {eq:boundprin15}

where g is a positive continuous function defined on $[a, \infty)$ and such that

$$\int_b^\infty \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} dt < \infty$$

for some $b \geq a$. If we want F to be positive, the expression in (33) may fail to achieve that on the whole line. So we define it on $[a, \infty)$ and then extend F to all of $\mathbb{R}$ in such a way that it becomes a strictly increasing C^2 function.

Let's now translate conditions (31) and (32) into conditions on the function g. To do so, we first compute the relevant quantities.

$$F'(s) = \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad ,$$

and therefore

$$F''(s) = -\frac{1}{2} \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{3}{2}} g(s) \quad .$$

Hence

$$\frac{|F'(t)|^2}{F(t)\,|F''(t)|} = 2\, \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} \quad ,$$

so condition (31) is equivalent to

$$\limsup_{t \to \infty} \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} < \infty \quad .$$

Moreover,

$$\frac{F'(t)^3 f(t)}{F(t)^4 |F''(t)|} = 2\, \frac{f(t)}{F(t)^4 g(t)} \quad ,$$

so condition (32) is equivalent to

$$\liminf_{t \to \infty} \frac{f(t)}{F(t)^4 g(t)} > 0 \quad .$$

Finally, we observe that since g is positive, if $s \leq t$, then

$$\left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} \geq \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad .$$

Therefore, for $t \geq b$,

$$\begin{aligned} F(t) - 1 &= \int_a^t \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} ds \\ &\geq \int_b^t \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} ds \\ &\geq \int_b^t \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} ds \\ &= (t-b) \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad . \end{aligned}$$

Hence

$$F(t) \geq (t-b) \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad .$$

This allows us to further estimate the quantity appearing in (31):

$$\begin{aligned} \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} &\leq \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{(t-b) \left(\int_a^t g(\tau)\, d\tau \right)^{-1/2} g(t)} \\ &= \frac{\int_a^t g(\tau)\, d\tau}{(t-b) g(t)} \\ &\sim \frac{\int_a^t g(\tau)\, d\tau}{t\, g(t)} \quad . \end{aligned}$$

We have thus proved the following theorem. todo: commento prof

{th:boundprin}

Theorem G.2. *Let M be a Riemannian manifold whose Ricci curvature is bounded from below, and let $u \in C^2(M)$ satisfy*

$$\Delta u \geq f(u) \quad .$$

A sufficient condition for u to be bounded from above on M is the existence of a continuous positive function g, defined on an interval $[a, \infty)$, such that

- $\displaystyle\int_b^\infty \left(\int_a^t g(\tau)\, d\tau \right)^{-1/2} dt < \infty$ *for some* $b \geq a$,
- $\displaystyle\limsup_{t \to \infty} \frac{\int_a^t g(\tau)\, d\tau}{t\, g(t)} < \infty$,
- $\displaystyle\liminf_{t \to \infty} \frac{f(t)}{g(t)} > 0$.

G.2 The uniqueness theorem

In order to prove uniqueness, we pass to an auxiliary formulation of the problem. We first prove uniqueness in this new setting, and then show that this implies

uniqueness for the original equation.
Recall that we are looking for a function w such that the metric g can be written as

$$g = e^{2w} ds_p^2,$$

where ds_p^2 is the Poincaré metric on $\mathbb{D}$. Since ds_p^2 is itself conformal to the Euclidean metric $|dz|^2$, with conformal factor

$$\frac{4}{(1-|z|^2)^2},$$

it follows that g is also conformal to $|dz|^2$. In fact, we can write

$$g = \frac{4e^{2w}}{(1-|z|^2)^2}|dz|^2.$$

If we now express the conformal factor in the form $e^{2\tilde{w}}$, we find a transformation between w and $\tilde{w}$,

$$\tilde{w} = w + \log\left(\frac{2}{1-|z|^2}\right).$$

This lets us write the metric g as

$$g = e^{2\tilde{w}}|dz|^2.$$

Therefore w and $\tilde{w}$ describe the same metric, the former with respect to the Poincaré background metric and the latter with respect to the Euclidean one. We perform this transformation as it will be convenient to study the corresponding equation in the Euclidean setting first, where the relevant operator is the standard Laplacian Δ. Once we have proved uniqueness there, we will translate the result back to the original formulation.

Theorem G.3. *Let h be a continuous function defined on a domain $\Omega \subset \mathbb{C}$. Let $\tilde{w}$ be a solution of*

$$\Delta\tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \tag{34}$$ {eq:unithm}

in Ω, and assume that

$$\Delta\tilde{w} \geq 0 \quad ,$$

and that $e^{2\tilde{w}}|dz|^2$ is a complete metric on Ω. Then any other solution v of the equation satisfies $v \leq \tilde{w}$.

Proof. Let us define $\eta = v - \tilde{w}$. We need to prove that $\eta \leq 0$.
We have

$$\begin{aligned}\Delta\eta &= \Delta v - \Delta\tilde{w} \\ &\overset{[(34)]}{=} e^{2v} - h^2 e^{-2v} - e^{2\tilde{w}} + h^2 e^{-2\tilde{w}} \quad .\end{aligned} \tag{35}$$ {eq:unithm2}

Let's now try to explicit $\tilde{w}$ and η in the RHS of (35),

$$e^{2v} = e^{2(\eta+\tilde{w})} = e^{2\eta}e^{2\tilde{w}} \quad .$$

Let's define the quantity $|\mu|^2 = h^2 e^{-4\tilde{w}}$, we have then

$$h^2 e^{-2v} = h^2 e^{-2(\eta+\tilde{w})} = h^2 e^{-2\tilde{w}} e^{-2\eta} = |\mu|^2 e^{2\tilde{w}} e^{-2\eta},$$

and

$$h^2 e^{-2\tilde{w}} = \left(h^2 e^{-4\tilde{w}}\right) e^{2\tilde{w}} = |\mu|^2 e^{2\tilde{w}} \quad .$$

So, by substituting what we just found in (35), we get

$$\begin{aligned} \Delta\eta &= e^{2\tilde{w}} e^{2\eta} - |\mu|^2 e^{2\tilde{w}} e^{-2\eta} - e^{2\tilde{w}} + |\mu|^2 e^{2\tilde{w}} \\ &= e^{2\tilde{w}} (e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2) \quad . \end{aligned} \tag{36}$$ {eq:unithm3}

The Laplacian with respect to the metric $g = e^{2\tilde{w}}|dz|^2$ is simply $e^{-2\tilde{w}}\Delta$. So in (36) we obtain

$$\Delta_g \eta = e^{-2\tilde{w}} \Delta\eta = e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2 \quad . \tag{37}$$ {eq:unithm5}

Let us now observe that $|\mu|^2 \leq 1$. Indeed, by (34),

$$e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \quad ,$$

hence

$$e^{2\tilde{w}} (1 - h^2 e^{-4\tilde{w}}) \geq 0 \quad ,$$

that is,

$$1 - |\mu|^2 \geq 0 \quad ,$$

so $|\mu|^2 \leq 1$.
Therefore, from (37) we get

$$\begin{aligned} \Delta_g \eta &= e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2 \\ &\geq e^{2\eta} - e^{-2\eta} - 1 =: f(\eta) \quad . \end{aligned}$$

We now apply Theorem G.2 to the function η on the complete simply connected Riemannian surface (Ω, g). For the auxiliary function appearing in the theorem, we may take

$$\gamma(t) = t^2$$

on $[1, \infty)$. A direct computation shows that γ satisfies the required assumptions, since

$$\int_b^\infty \left(\int_1^t \tau^2 \, d\tau \right)^{-1/2} dt < \infty, \qquad \limsup_{t\to\infty} \frac{\int_1^t \tau^2 \, d\tau}{t^3} < \infty,$$

and

$$\liminf_{t\to\infty} \frac{f(t)}{t^2} = \liminf_{t\to\infty} \frac{e^{2t} - e^{-2t} - 1}{t^2} > 0 \quad .$$

It remains only to check that the Ricci curvature of g is bounded from below which, since we are on a surface, is the same as the Gaussian curvature. For the metric

$$g = e^{2\tilde{w}} |dz|^2 \quad ,$$

the Gaussian curvature is

$$\begin{aligned} K(g) &= -e^{-2\tilde{w}}\Delta\tilde{w} \\ &= -e^{-2\tilde{w}}\left(e^{2\tilde{w}} - h^2 e^{-2\tilde{w}}\right) \\ &= -1 + h^2 e^{-4\tilde{w}} \\ &= -1 + |\mu|^2 \geq -1 \quad . \end{aligned} \tag{38}$$

{eq:unithm10}

Since $|\mu|^2 \leq 1$, it follows that

$$-1 \leq K(g) \leq 0 \quad ,$$

and in particular, looking at (38), the curvature is bounded from below. Therefore we can apply Theorem G.2 and conclude that η is bounded from above. Now we can also apply Theorem G.1 to η, obtaining a sequence $(z_k)_k \subset \Omega$ such that

$$\eta(z_k) \to \bar{\eta} = \sup_{\Omega} \eta, \qquad \limsup_{k\to\infty} \Delta_g \eta(z_k) \leq 0 \quad .$$

Since $|\mu|^2$ is continuous and bounded, by passing to a subsequence if necessary we may also assume that

$$|\mu|^2(z_k) \to a$$

for some $a \leq 1$. Taking limits in (37), we obtain

$$\begin{aligned} 0 &\geq \limsup_{k\to\infty} \Delta_g \eta(z_k) \\ &= \lim_{k\to\infty} \left(e^{2\eta(z_k)} - |\mu|^2(z_k) e^{-2\eta(z_k)} - 1 + |\mu|^2(z_k)\right) \\ &= e^{2\bar{\eta}} - a e^{-2\bar{\eta}} - 1 + a \\ &=: \varphi(\bar{\eta}) \quad . \end{aligned} \tag{39}$$

{eq:unithm4}

We have defined in the last expression the function $\varphi(s) = e^{2s} - ae^{-2s} - 1 + a$, which has strictly positive derivative $\varphi'(s) = 2e^{2s} + 2ae^{-2s} > 0$, so that it's strictly increasing. Moreover $\varphi(0) = 0$, hence (39) implies

$$\bar{\eta} \leq 0, \qquad \text{therefore, on } \Omega, \qquad \eta \leq 0 \quad .$$

This finally implies $v \leq \tilde{w}$. □

{cor:uni1}

Corollary G.1. *Let h be a continuous function defined on a domain $\Omega \subset \mathbb{C}$ and let $\tilde{w}_1, \tilde{w}_2$ two solutions of*

$$\Delta\tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \tag{40}$$

in Ω such that $e^{2\tilde{w}_i}$ are complete metrics on Ω. Then $\tilde{w}_1 \equiv \tilde{w}_2$.

Proof. Just observe that for the previous theorem, $\tilde{w}_1 \leq \tilde{w}_2$ and $\tilde{w}_2 \leq \tilde{w}_1$. □

{cor:uni2}

Corollary G.2. *There is at most one solution of the equation* (13) *such that the associated metric is complete.*

Proof. Let w_1, w_2 be two solutions of

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1$$

such that (13) holds and the metrics $e^{2w_1} ds_p^2$ and $e^{2w_2} ds_p^2$ are complete.

As we mentioned in the introduction, we can translate this problem into one involving the Euclidean Laplacian by means of the following transformation: for $i = 1, 2$, define

$$\tilde{w}_i = w_i + \log\left(\frac{2}{1-|z|^2}\right) \quad . \tag{41}$$ {eq:unithm6}

Since

$$ds_p^2 = \frac{4}{(1-|z|^2)^2}|dz|^2,$$

we have

$$e^{2\tilde{w}_i}|dz|^2 = e^{2w_i}\frac{4}{(1-|z|^2)^2}|dz|^2 = e^{2w_i} ds_p^2.$$

Hence $e^{2\tilde{w}_i}|dz|^2$ is complete.

Let us now show that each $\tilde{w}_i$ solves equation (13) and satisfies (14). First we recall that

$$\Delta_p = \frac{(1-|z|^2)^2}{4}\Delta$$

and observe that

$$\Delta \log\left(\frac{2}{1-|z|^2}\right) = \frac{4}{(1-|z|^2)^2} \quad .$$

So it follows that

$$\begin{aligned}
\Delta\tilde{w}_i &= \Delta w_i + \Delta \log\left(\frac{2}{1-|z|^2}\right) \\
&= \frac{4}{(1-|z|^2)^2}\Delta_p w_i + \frac{4}{(1-|z|^2)^2} \\
&= \frac{4}{(1-|z|^2)^2}\left(e^{2w_i} - q^2 e^{-2w_i} - 1\right) + \frac{4}{(1-|z|^2)^2} \\
&= \frac{4}{(1-|z|^2)^2}e^{2w_i} - \frac{4}{(1-|z|^2)^2}q^2 e^{-2w_i}.
\end{aligned} \tag{42}$$ {eq:unithm7}

Now, from (41) we get

$$\frac{4}{(1-|z|^2)^2}e^{2w_i} = e^{2\tilde{w}_i}, \qquad e^{-2w_i} = \frac{4}{(1-|z|^2)^2}e^{-2\tilde{w}_i}.$$

Therefore, substituting in (42), we obtain

$$\Delta\tilde{w}_i = e^{2\tilde{w}_i} - \left(\frac{4q}{(1-|z|^2)^2}\right)^2 e^{-2\tilde{w}_i} \quad . \tag{43}$$ {eq:unithm8}

Define

$$h(z) = \frac{4}{(1-|z|^2)^2}q(z).$$

Then h is continuous on $\mathbb{D}$, and (43) becomes

$$\Delta \tilde{w}_i = e^{2\tilde{w}_i} - h^2 e^{-2\tilde{w}_i} \quad . \tag{44}$$ {eq:unithm9}

Moreover, since (14) holds for w_i, we have

$$e^{2w_i} - q^2 e^{-2w_i} > 0,$$

and

$$\Delta \tilde{w}_i > 0.$$

This means that $\tilde{w}_1$ and $\tilde{w}_2$ are two solutions of (13) and (14) with complete metrics $e^{2\tilde{w}_i}|dz|^2$. Finally we apply corollary G.1 to conclude that $\tilde{w}_1 = \tilde{w}_2$. Hence, by (41), we conclude that

$$w_1 = w_2.$$

□

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